

Week 4 Journal

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PROJECT 3

Wednesday, September 24, 2025

- Today we did the Oscilloscope ppt in class I literally knew what nonblocking is but I froze when asked by my professor.
- I started working on hardware after getting new dc motor, small breadboard for the lab.
- I tested both motors and placed the breadboard on the robot.
- Assigned Nikunj to take care of readme from GitHub.
- I understood how the battery pack placed beneath the robot needs to be used as a power source.

Thursday, September 25, 2025

- Today after telecom class I setup the whole thing by mounting breadboard and pi on the chassis.
- I figured out how to do the wiring.
- Soldered jumpers to Dc motor and battery switch.
- I did soldering because it was necessary as the wires were not connecting securely.
- First I tried to stay away from soldering but it did not support, it was fluctuating so much I thought motor was defective. After soldering everything started working fine.
- This video helped me a lot. https://youtu.be/T7fudh_BbIA?si=qnsHOWBNmbSBUMBo

Friday, September 26, 2025

- I worked on the coding part
- Bought a power bank, 9v battery and some AA batteries.
- I made one testing code that was no inputs just 2s forward and 2s reverse.
- I made another code that accepted inputs WASD.
- Another one for controlling speed(PWM) + WASD.
- TinkerCad does not support pi so I tried downloading alternative like fritzing.

Monday, September 29, 2025

- Today I tested the final code and wiring for the project.
- It worked fine but for like 5 minutes then the motors made a sound but did not work.
- I tried troubleshooting and changing the IC.
- Tested the motors separately and they worked fine.

- Tried new code but had the same problem.

Tuesday, September 30, 2025

- Today I tested if EN1,2 are working properly by placing LED with resistor.
- Also noticed the duty cycle with the help of it.
- Worked till late midnight and figured out my wiring was a step down.
- Fixed the wiring and made the setup secure by using double tape.
- Made the car run a little bit to test.